VWO Login Dashboard - Test Plan

Introduction

The goal of this project is to test the VWO Login Dashboard. This includes ensuring secure, accessible, and high-performance authentication for users. The dashboard supports new, returning, and error-recovery flows, targeting a seamlessly unified experience meeting GDPR, Enterprise Security, and WCAG accessibility standards.

Scope (Tested & Not Tested)

Tested (In-Scope)

- **Core Authentication:** Secure login form, basic validation, error handling, password reset, loading states, email validation.
- **Enhanced UX:** Mobile optimization, responsive design, visual feedback for errors, theme support (Light and Dark mode options).
- **Enterprise Features:** SSO integration capabilities (SAML, OAuth), optional social login (Google, Microsoft).
- **Security Specifics:** Encrypted password storage, end-to-end HTTPS encryption, session security, rate limiting (protection against brute force).
- **Performance:** Sub-2-second page load time, asset optimization, high availability/ concurrency support.
- **Accessibility:** WCAG 2.1 AA compliance, screen reader support (ARIA labels), keyboard navigation, high contrast mode.

Not Tested (Out-of-Scope)

- **Marketing Tools Integration:** [NOT SPECIFIED] Detailed specs on integration with marketing and customer onboarding platforms.
- **Analytics & A/B Testing:** [NOT SPECIFIED] The PRD mentions use of "VWO's own platform" for A/B testing, but clear requirements on the data endpoints are not provided.
- **Advanced Features:** Biometric authentication and Adaptive Authentication are slated for "Future Enhancements" and are excluded from current testing.

Test Strategy/Approach

We will leverage Playwright for our automated testing framework, given its capabilities in modern web application testing across diverse browser engines.

- **Functional Testing:** Automated via Playwright focused on user journeys (New User, Returning User, Error Recovery).
- **Performance Testing:** Validate page load speeds ($\leq$ 2s) and asset optimization via browser profiling tools.
- **Security Testing:** API intercept testing and negative tests for brute force (rate limiting validation).
- **Accessibility Testing:** Automated Axe-core integration within Playwright to check WCAG compliance and keyboard navigation.

Test Environment

- **Automation Framework:** Playwright (Node.js/TypeScript).
- **Browsers:** Chromium, WebKit, Firefox (Desktop and Mobile viewports).
- **Network:** Specific tests simulated over standard connections to enforce the $\leq$ 2-second SLA.
- **OS:** Cross-platform (Windows, macOS, Linux).

Test Criteria

- **Entry Criteria:**
  - Login UI and authentication endpoints are deployed to the QA environment.
  - Test data (users, SSO accounts) is provisioned.
- **Exit/Pass-Fail Criteria:**
  - 100% test execution with a minimum of 95% pass rate for Core Authentication.
  - Zero critical security vulnerabilities and 100% compliance with security audit requirements.
  - Page load time strictly $\leq$ 2 seconds for successful validations.

Resources & Roles

- **QA Automation Engineer:** Develops Playwright test scripts, maintains test cases.
- **Security Tester:** Conducts specific tests covering rate limiting and HTTPS enforcement.
- **Performance Tester:** Analyzes CDN and asset load times.
- **[NOT SPECIFIED] :** Specific team sizes and allocated individual names are absent from the PRD.

Schedule & Milestones

- **Phase 1: Q1** - Core Authentication (Secure login form, basic validation, password reset).
- **Phase 2: Q2** - Enhanced UX (Mobile optimization, accessibility features, advanced validation).
- **Phase 3: Q3** - Enterprise Features (SSO implementation, advanced security, monitoring).

Risk & Contingency

- **Security Risks:** Risk of unauthorized access. *Mitigation:* Implement real-time monitoring and security patch deployment.
- **Performance Risks:** System failure under high simultaneous logins. *Mitigation:* Load testing under varied conditions and auto-scaling infrastructure verification.
- **Missing Information ([NOT SPECIFIED]):** Gaps in requirements related to Analytics endpoints and Marketing tool integration. *Mitigation:* Escalate doubts to Product Management.

Deliverables

- Automated Playwright Test Suite (Code repository).
- Test Execution Report (HTML/PDF formats).
- Performance & Accessibility Audit Summaries.

Doubts & Gaps Raised:

1. **Analytics Endpoints:** The PRD mentions tracking login success/failure for optimization. What are the specific analytics events and payload structures expected?
2. **SSO Provider specifics:** Which specific "enterprise authentication protocols" other than SAML/OAuth are expected in Phase 3?
3. **Enterprise Security Policies:** What specific "enterprise security policies and audit requirements" are targeted? [Not Specified]